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Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Engg.) Final Examination-2023
CSE 2103: Data Structure and Algorithms-I (3 Credits)

Time: 3 Hours

Full Marks: 70

INSTRUCTIONS

- a. Answer **05** out of **07** questions
- b. Figures in the margin indicate **full marks**

Question-1

- a. Differentiate between linear and non-linear data structures, and describe the key factors used to evaluate the performance of an algorithm. 04
- b. For each example below, there are two algorithms solving the same problem. Given the asymptotic runtimes for each, is one of the algorithms guaranteed to be faster? If so, which? And if neither is always faster, explain why. Assume the algorithms have very large input (i.e. N is very large). 06
 - I. Algorithm 1: $\Theta(N)$ Algorithm 2: $\Theta(N^2)$
 - II. Algorithm 1: $\Omega(N)$ Algorithm 2: $\Omega(N^2)$
 - III. Algorithm 1: $O(N)$ Algorithm 2: $O(N^2)$
 - IV. Algorithm 1: $\Theta(N^2)$ Algorithm 2: $O(\log N)$
 - V. Algorithm 1: $O(N \log N)$ Algorithm 2: $\Omega(N \log N)$

- c. Define recursive function. Consider the following recursive function. 04

```
public void andslam(int N) {  
    if (N > 0) {  
        for (int i = 0; i < N; i += 1) {  
            for (int j = 1; j < 1024; j *= 2) {  
                System.out.println(i + j);  
            }  
        }  
        andslam(N / 2);  
    }  
}
```

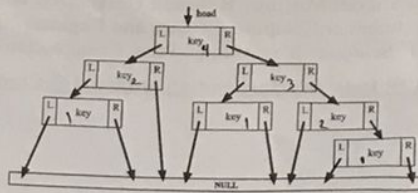
For the given recursive functions, provide asymptotic bounds for the best case and worst case runtimes in $\Theta(\cdot)$ notation with proper explanations. Give the runtime in terms of N .

Question-2

- a. Describe the method for identifying the middle element of a singly linked list with an unknown number of elements in just one iteration through the entire list. Following that, discuss why a linked list is not an appropriate data structure for executing a Binary Search. 05
- b. A Node in a linked structure contains three elements: an integer *key* and two pointers, *L* and *R*, pointing to other Nodes. Given the figure showing the current state of the linked structure, write the *key* values of each Node after executing *func(head)*. The function *func* is defined as follows. 05

```
int func (Node *temp) {  
    if (temp == NULL) return 0;  
    int x = func(temp->L);  
    int y = func(temp->R);  
    temp->key = 1 + max(x, y); //max(x,y) returns the maximum value between x and y  
    return temp->key;  
}
```

You can assume that the *head* pointer has been declared globally.



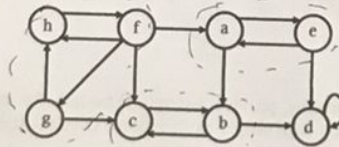
- c. For the following recurrences, give an expression for the runtime $T(n)$ if the recurrence can be solved with the Master Theorem. Otherwise, indicate that the Master Theorem does not apply. 04
- I. $T(n) = 3T(n/2) + n$ II. $T(n) = 64T(n/8) - n^2 \log n$

Question-3

- a. Compare the performance of using an adjacency list versus an adjacency matrix for implementing Breadth-First Search (BFS) on the following types of graphs: 03
- I. Complete Graph
II. Cycle Graph
- b. Consider the following table that outlines course prerequisites. A prerequisite must be completed before enrolling in a course. 05

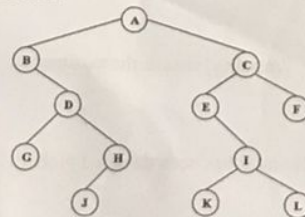
Course	Prerequisite
CSE 2203	CSE 1105
CSE 2220	CSE 2205, CSE 2203
CSE 2205	CSE 1105
CSE 2215	CSE 2203
CSE 2279	CSE 1169

- c. Identify the strongly connected components within the given graph. Utilize an appropriate algorithm to develop your solution, providing a step-by-step simulation of the process. Finally, illustrate the resulting component graph. 06



Question-4

- a. A graph with n vertices and $n-1$ edges is a tree. Express your opinion on this statement, and provide necessary examples to support your answer. 02
- b. Discuss how a Binary Search Tree can be used for sorting a sequence of integers. 02
- c. A finite tree T has at least one vertex v of degree 4, and at least one vertex w of degree 3. Prove that T has at least 5 leaves. 04
- d. Given the following binary tree, write the pre-order, in-order, and post-order traversals of its vertices and then discuss how to check if a given binary tree is BST or not. Consider the node A as the root of the tree. 06



Question-5

- a. Describe the operations of a Disjoint Set, and then propose a technique to improve the running time of the following function *findset* for finding the representative of an element within the disjoint set. You may assume that the elements of the disjoint set are integers, represented as the index of the *parent* array. The function is written in C language. 04

```
int findset(int x){
    if(x == parent[x]) return x;
    return findset(parent[x]);
}
```

- b. Consider the following matrix *G* that represents an undirected weighted graph. In the matrix, $G[u][v] = w$ represents that there is an edge between *u* and *v* and the weight of the edge is *w* where *w* is a positive integer. $G[u][v] = 0$ indicates that there is no edge between *u* and *v*. 10

G	A	B	C	D	E	F
A	0	8	1	3	0	0
B	8	0	0	3	0	0
C	1	0	0	4	3	1
D	3	3	4	0	2	2
E	0	0	3	2	0	1
F	0	0	1	2	1	0

Demonstrate all the necessary steps to calculate the minimum spanning tree of the given graph *G* using Kruskal's algorithm. At each step of the algorithm, illustrate the relevant operations of the disjoint set, utilizing the union by size heuristic.

Question-6

- a. Explain how using a Heap improves the performance of Prim's algorithm compared to using an array to manage the sequence of vertices with the lowest key. 04
- b. When a sequence consists of *n* values that are all equal and need to be inserted into a heap, demonstrate that the overall time complexity for inserting these values is $O(n)$ rather than $O(n \log n)$. Provide an example to illustrate your explanation. 04
- c. Consider a two dimensional coordinate system where i^{th} point is represented as $P_i(x_i, y_i)$. Distance between 2 points P_i and P_j is measured as $|x_i - x_j| + |y_i - y_j|$. 06

Now a list of 4 points are given in Table-1.

i	Point	x_i	y_i
1	P_1	2	-10
2	P_2	4	0
3	P_3	2	6
4	P_4	0	0

Illustrate a weighted graph from the scenario described above where points are labeled as vertices and weight of an edge is calculated by the distance of two of its end vertices.

Construct a minimum spanning tree of the graph using Prim's algorithm showing all the necessary steps and mention the cost of the tree as well.

Question-7

- a. Determine the maximum number of nodes in a binary tree of height *k*. 02
- b. Explain when a doubly linked list is more efficient than a singly linked list. 02
- c. Describe how a complete binary tree can be represented using both an array and a linked data structure. 03
- d. Determine the longest common subsequence between the two strings provided below. 07
- ABCBADAB
 - BDCABA
- Present the relevant matrix and detail all necessary steps to support your solution.